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Explainable AI LAB – 03

Title: SHAP (SHapley Additive Explanations) for Feature Importance

Objective

To understand and implement Explainable Artificial Intelligence (XAI) using **SHAP (SHapley Additive Explanations)** in order to interpret **local and global feature contributions** in predictions made by black-box machine learning models on tabular, text, or image datasets.

Learning Outcomes

After completing this experiment, students will be able to:

1.Explain the need for explainability in machine learning models
2.Understand the concept of Shapley values from game theory
3. Implement SHAP for tabular, text, and image data
4. Interpret local and global feature importance using SHAP plots
5. Analyze model behavior, bias, and feature interactions
6. Compare SHAP explanations with human intuition

Tools & Libraries

- Python 3.x
- NumPy, Pandas
- Scikit-learn / TensorFlow / PyTorch
- SHAP
- Matplotlib / Seaborn (for visualization)

Background / Theory

SHAP (SHapley Additive Explanations) is an Explainable AI technique based on **Shapley values from cooperative game theory**. It explains model predictions by **fairly distributing the contribution of each feature** to the final prediction.

SHAP expresses a prediction as:

Prediction = Baseline + Sum of Feature Contributions

Unlike LIME, SHAP provides:

- **Local explanations** (individual predictions)
- **Global explanations** (overall feature importance)

SHAP satisfies three desirable properties:

- **Local accuracy**

- **Consistency**
- **Missingness**

📌 This makes SHAP *theoretically stronger* than LIME.

Key Characteristics of SHAP:

Local and Global:
Explains individual predictions (local explanations) as well as overall model behavior (global feature importance)

• **Additive and Consistent:**
Represents a prediction as a sum of feature contributions, ensuring consistency and fairness in attribution

• **Model-Agnostic (and Model-Specific):**
Can be applied to any machine learning model, with optimized variants available for specific models (e.g., TreeSHAP, DeepSHAP)

• **Theoretically Grounded:**
Based on Shapley values from cooperative game theory, satisfying properties of local accuracy, consistency, and missingness

• **Feature Contribution Oriented:**
Quantifies how much each feature increases or decreases the prediction relative to a baseline value

Problem Statement

Given a trained black-box machine learning model, explain why the model made a particular prediction using SHAP by quantifying the contribution of each feature, and analyze whether the explanation is logical, meaningful, consistent across instances, and unbiased.

Group Allocation

The class is divided into **20 groups**.
Each group will work with **one dataset and one black-box model**, selected from **tabular, text, or image domains**, and apply **SHAP (SHapley Additive Explanations)** to interpret model predictions.

Group-Wise Dataset & Task Allocation

Tabular Data Groups

These are best suited for **classical machine learning models** and for understanding **local and global feature importance using SHAP**.

1. Titanic Survival Dataset (Kaggle)

Task: Predict survival
Why good: Simple, interpretable features (age, sex, class)
→ Ideal for demonstrating **SHAP feature contributions and global importance**

2. UCI Adult Income Dataset

Task: Predict income >50K
Why good: Fairness, bias detection, and explainability use-case
→ Effective for analyzing **SHAP values related to sensitive attributes**

3. UCI Heart Disease Dataset

Task: Predict presence of heart disease
Why good: Healthcare relevance + ethical AI perspective
→ SHAP helps explain medically significant features

4. UCI Breast Cancer Wisconsin Dataset

Task: Benign vs malignant tumor classification
Why good: Medical dataset with strong feature importance patterns
→ Suitable for **SHAP summary and waterfall plots**

5. Loan Approval Dataset (Kaggle / UCI)

Task: Predict loan approval
Why good: Real-world decision-making scenario
→ SHAP explains approval/rejection decisions transparently

6. German Credit Dataset (UCI)

Task: Credit risk prediction
Why good: Bias and fairness discussions
→ SHAP reveals contribution of socio-economic features

7. Diabetes Dataset (Pima Indians)

Task: Diabetes prediction
Why good: Healthcare + tabular SHAP explanations
→ Useful for interpreting clinical feature influence

8. Wine Quality Dataset (UCI)

Task: Predict wine quality score
Why good: Regression task
→ Demonstrates **SHAP values for regression models**

9. Bank Marketing Dataset (UCI)

Task: Predict term deposit subscription
Why good: Business analytics + interpretability
→ SHAP highlights marketing feature impact

10. Student Performance Dataset (UCI)

Task: Predict student grades
Why good: Education-relevant and interpretable features
→ SHAP explains academic performance predictors

Text Datasets

These datasets are suitable for showing **which words or tokens influence predictions**, using **SHAP explanations for NLP models**.

11. IMDB Movie Reviews

Task: Sentiment analysis
Why good: SHAP highlights influential words contributing to sentiment

12. SMS Spam Collection (UCI)

Task: Spam vs ham classification
Why good: Very intuitive word-level SHAP explanations

13. 20 Newsgroups Dataset

Task: Topic classification
Why good: Multi-class NLP explanations using SHAP

14. Amazon Product Reviews

Task: Sentiment or rating prediction
Why good: Real-world NLP explainability
→ SHAP explains review-based predictions

15. Yelp Reviews Dataset

Task: Sentiment / star rating prediction
Why good: Business-oriented NLP dataset
→ SHAP reveals important opinion words

Image Datasets

These datasets are engaging for students and are used to visualize **important image regions influencing predictions** using SHAP.

16. MNIST (Handwritten Digits)

Task: Digit classification
Why good: Simple CNN model
→ SHAP highlights influential pixel regions

17. Fashion-MNIST

Task: Clothing classification
Why good: More visually rich than MNIST
→ SHAP explains region-level importance

18. CIFAR-10

Task: Object classification
Why good: Color images with complex patterns
→ SHAP visualizations show feature influence in images

19. Cats vs Dogs (Kaggle)

Task: Binary image classification
Why good: Intuitive visual interpretation
→ SHAP clearly highlights relevant image regions

20. Caltech-101

Task: Multi-class object recognition
Why good: Diverse image categories
→ Suitable for advanced SHAP image explanations

Experiment Procedure

Each group must follow the steps below:

Step 1: Dataset Preparation

• Load and preprocess the dataset
• Handle missing values, encoding, and normalization as required
• Split the dataset into **training and testing sets**

Step 2: Train a Black-Box Model

Choose **one** of the following models:

• Random Forest
• XGBoost
• Neural Network (MLP)
• Convolutional Neural Network (for image datasets)

Train the selected model and record appropriate **performance metrics** such as accuracy, precision, recall, F1-score, or loss.

Step 3: Select Test Instances

• Select **3–5 test instances** for explanation
• Ensure diversity in predictions (correct predictions, incorrect predictions, positive/negative classes, and edge cases)

Step 4: Apply SHAP

• Use the appropriate **SHAP explainer** based on the model and data type:

- `shap.Explainer` (model-agnostic)
- `TreeExplainer` (for tree-based models)
- `DeepExplainer` (for neural networks)

• Compute **SHAP values** for the selected test instances
• Analyze how each feature, word, or image region contributes to the model's prediction

Step 5: Visualization

• Generate SHAP visualizations such as:

- Summary (Beeswarm) plots
- Bar plots (global feature importance)
- Waterfall plots (single prediction explanation)
- Force plots
- Image or text explanations (where applicable)

• Save screenshots or plots for **report submission**

Analysis Questions (Mandatory)

Each group must answer the following questions for **every test instance** explained using **SHAP**:

Q1. Which features / words / image regions mattered most?

• Identify the **top contributing features** based on SHAP values
• Clearly mention **positive and negative contributions** relative to the baseline prediction
• Distinguish between features that **increase** and **decrease** the model output

Q2. Do the explanations make human sense?

• Are the features identified by SHAP **logically relevant** to the prediction task?
• Compare the model's reasoning (as shown by SHAP) with **human intuition or domain knowledge**

Q3. Are there signs of bias or spurious correlations?

• Is the model relying heavily on **irrelevant or sensitive attributes**?
• Are there indications of **unfair influence** (e.g., gender, age, keywords, background pixels)?
• Discuss how SHAP helps in identifying such biases

Q4. Local vs Global Insight (Reflection)

• Compare **local SHAP explanations** across multiple test instances
• Analyze **global feature importance** using SHAP summary or bar plots
• Do explanations remain consistent across instances?
• What does this indicate about the **stability and behavior** of the model?

Expected Outcome

• Students should successfully generate **local and global explanations** for black-box predictions using SHAP
• Understand the **strengths and limitations of SHAP** as an Explainable AI technique
• Develop **critical thinking** regarding AI transparency, fairness, and trustworthiness
• Gain insight into how feature contributions influence model decisions

Viva Voce Questions (Sample)

1. What are **Shapley values**, and how are they used in SHAP?
2. Why is SHAP considered **theoretically stronger** than LIME?

3. What is the **baseline prediction** in SHAP?
4. Difference between **local and global explanations** in SHAP
5. Why can SHAP be **computationally expensive**?
6. In what situations should SHAP explanations be interpreted carefully?

Result

(Screenshots of **code, SHAP plots, and output** to be uploaded by the student.)

Conclusion

(To be written by the student, reflecting on insights gained from SHAP-based explanations.)

Evaluation Rubric

Rubric Breakdown

Criteria	Description	Marks
1. Model Implementation	Correct selection, training, and functioning of a black-box model (RF / XGBoost / NN / CNN)	2
2. Correct Use of SHAP	Appropriate SHAP explainer used (model-agnostic / TreeSHAP / DeepSHAP) with correct configuration and execution	2
3. Quality of Explanation	Clear identification of important features / words / image regions influencing the prediction using SHAP	2
4. Interpretation & Reasoning	Ability to explain whether the SHAP output makes human sense and relate it to domain logic	2
5. Bias & Critical Analysis	Identification of bias, spurious correlations, or limitations in SHAP explanations	1
6. Presentation & Documentation	Neat report, correct screenshots/plots, clarity of answers, and proper structure	1
Total		**10**

Focus evaluation more on **interpretation quality** rather than just code execution. Students should demonstrate **thinking and reasoning**, not just running SHAP.

SHAP enables both **local explanations (individual predictions)** and **global explanations (overall feature importance)**, making it a powerful tool for understanding and trusting black-box machine learning models across data modalities.